

EXPERIMENT 4

HILL CIPHER

AIM

To implement the Hill Cipher substitution technique using Python.

ALGORITHM

1. Read the plaintext from the user.
2. Read the 3×3 key matrix.
3. Convert the plaintext into numerical values (A=0, B=1, ..., Z=25).
4. Multiply the plaintext vector with the key matrix modulo 26.
5. Convert the result back to characters to obtain the cipher text.
6. Display the encrypted text.

PYTHON CODE

```
key = [  
    [6, 24, 1],  
    [13, 16, 10],  
    [20, 17, 15]  
]  
  
text = input("Enter Plain Text (3 letters): ").upper()  
  
while len(text) < 3:  
    text += "X"  
  
text = text[:3]  
  
P = [ord(ch) - 65 for ch in text]  
  
C = []
```

```

for i in range(3):
    total = 0
    for j in range(3):
        total += key[i][j] * P[j]
    C.append(total % 26)

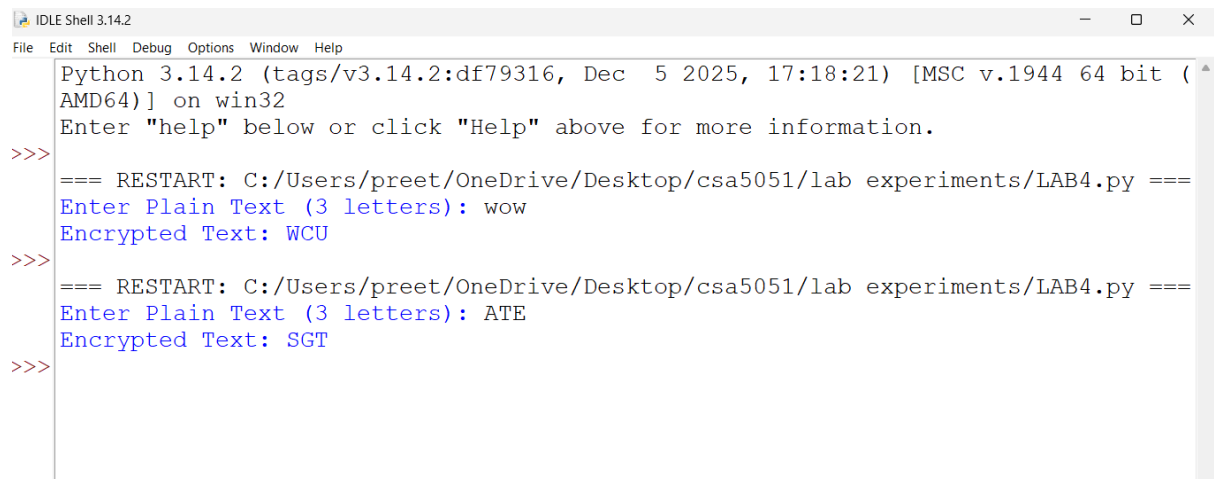
cipher = ""

for num in C:
    cipher += chr(num + 65)

print("Encrypted Text:", cipher)

```

OUTPUT



```

IDLE Shell 3.14.2
File Edit Shell Debug Options Window Help
Python 3.14.2 (tags/v3.14.2:df79316, Dec 5 2025, 17:18:21) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB4.py ===
Enter Plain Text (3 letters): wow
Encrypted Text: WCU
>>>
=== RESTART: C:/Users/preet/OneDrive/Desktop/csa5051/lab experiments/LAB4.py ===
Enter Plain Text (3 letters): ATE
Encrypted Text: SGT
>>>

```

RESULT

Thus the Hill Cipher substitution technique was implemented successfully using Python.